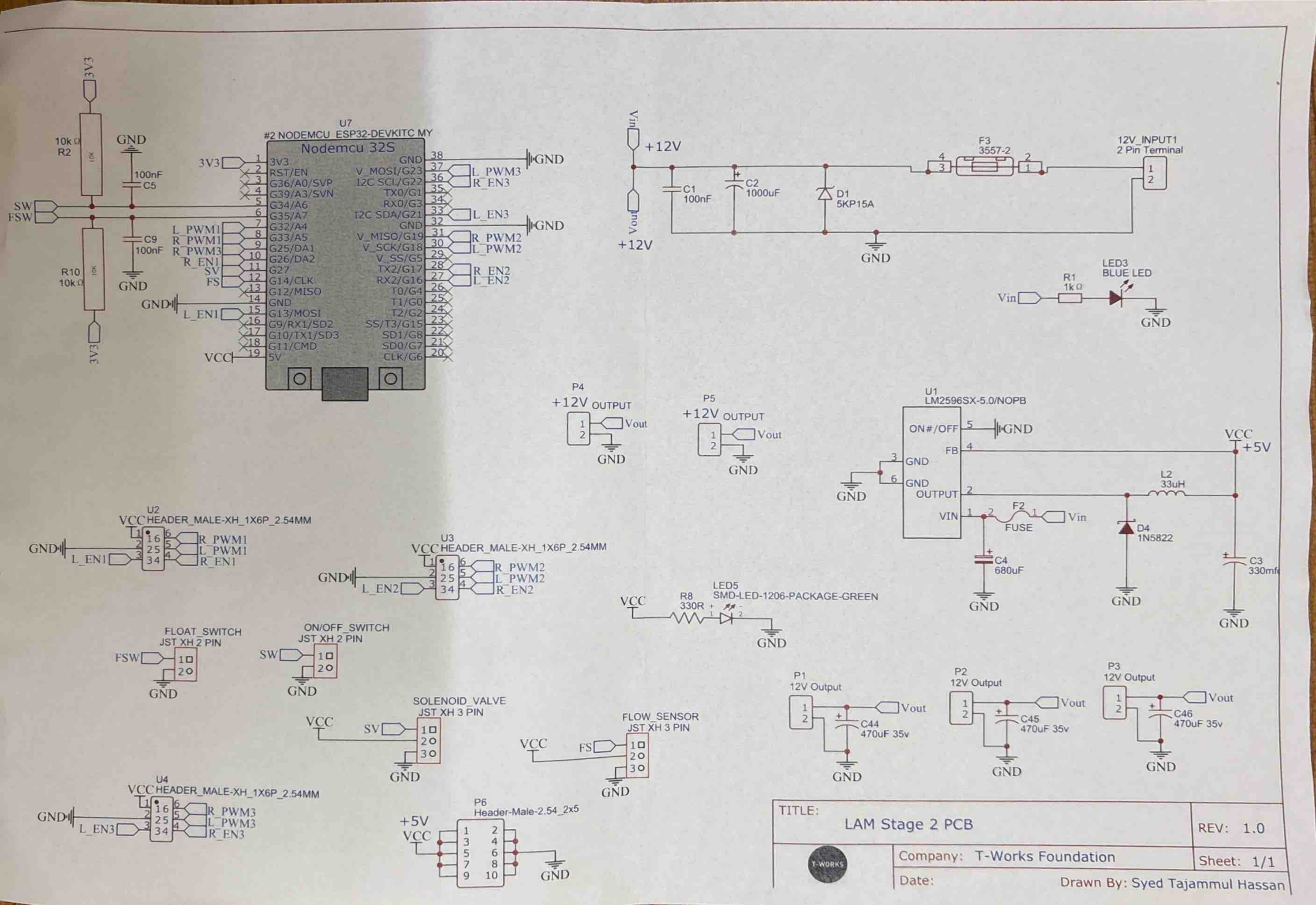

U7
#2 NODEMCU_ESP32-DEVKITC MY
Nodemcu 32S
3V3
GND
10k Ω
R2
100nF
C5
SW
FSW
C9
100nF
R10
10k Ω
L_PWM1
R_PWM1
R_PWM3
R_EN1
SV
FS
L_EN1
VCC
L_PWM3
R_EN3
L_EN3
R_PWM2
L_PWM2
R_EN2
L_EN2
Vin
+12V
Vout
C1
100nF
C2
1000uF
D1
5KP15A
F3
3557-2
12V_INPUT1
2 Pin Terminal
R1
1kΩ
LED3
BLUE LED
P4
+12V OUTPUT
P5
+12V OUTPUT
U1
LM2596SX-5.0/NOPB
ON#/OFF
FB
OUTPUT
VIN
L2
33uH
F2
FUSE
D4
1N5822
C4
680uF
C3
330mf
VCC
+5V
U2
HEADER_MALE-XH_1X6P_2.54MM
U3
HEADER_MALE-XH_1X6P_2.54MM
R8
330R
LED5
SMD-LED-1206-PACKAGE-GREEN
FLOAT_SWITCH
JST XH 2 PIN
ON/OFF_SWITCH
JST XH 2 PIN
SOLENOID_VALVE
JST XH 3 PIN
FLOW_SENSOR
JST XH 3 PIN
P1
12V Output
C44
470uF 35v
P2
12V Output
C45
470uF 35v
P3
12V Output
C46
470uF 35v
U4
HEADER_MALE-XH_1X6P_2.54MM
P6
Header-Male-2.54_2x5
TITLE:
LAM Stage 2 PCB
REV: 1.0
Company: T-Works Foundation
Sheet: 1/1
Date:
Drawn By: Syed Tajammul Hassan

Sr. No.	Section	Item / Specification	Qty per Team	Unit	TICK or Check list
		Mechanical			
1	Mechanical	PU tube – 8 mm OD × 5 mm ID	9	m	✓
2		PU tube – 10 mm OD × 8 mm ID {Blue Tube}	1	m	✓
3		Brass hose × ¼ inch	5	pcs	✓
4		Mini brass ball valve, ¼ inch F × F	3	pcs	✓
5		PU unequal union – 8 mm × 10 mm	6	pcs	✓
6		PU equal union – 8 mm	5	pcs	✓
7		PU female connector – ¼ inch × 8 mm	2	pcs	✓
8		PU male connector – ¼ inch × 8 mm	8	pcs	✓
9		PU equal tee – 8 mm	1	pc	✓
10		Tank 10 L	1	pcs	✓
11		Sump Tank 4.2L	1	pcs	✓
12		Flush Tank 4.2L	1	pcs	✓
13		Washer for ¼ inch hose	10	pcs	✓
		Electronics			
14	**Controller**	NodeMCU ESP32S / ESP32-WROOM-32 controller PCB	1	pc	✓
15	**Power**	12 VDC power supply / SMPS	1	pc	✓
16	**Motor Drive**	BTS7960 motor driver	3	pcs	✓
17	**Valve Drive**	12 VDC-compatible solenoid valve MOSFET driver module (D4184)	1	pc	✓
18	**Pump**	12 VDC water pump – agriculture knapsack sprayer double-motor type	3	pcs	✓
19	**Flow Control**	Flow sensor / YF-S401 flow transmitter	1	pc	✓
20	**Valve**	Brass diaphragm solenoid valve, ¼ inch SOV	1	pc	✓
21	**Level Sensor**	Horizontal float switch	1	pc	✓
22	**Switch**	ON/OFF tactile switch {TACK SWITCH}	1	pc	✓
		Wires			
23	Electrical/Wires	18 AWG red wire for Pump/motor wiring	10	m	✓
24		18 AWG black wire for Pump/motor wiring	10	m	✓
25		20 AWG red wire for Sensors/solenoid	5	m	✓
26		20 AWG black wire for Sensors/solenoid	5	m	✓
27		3-core wire for Flow sensor	5	m	✓
28		3-pin JST wire, 2.54 mm for Sensor/solenoid driver interface	1	pc	✓
29		2-core wire for Float switch	5	m	✓
30		2-pin JST wire, 2.54 mm for Float switch interface	1	pc	✓
31		3-pin JST wire, 2.54 mm for Flow sensor interface	1	pc	✓
32		2-pin JST wire, 2.54 mm for ON/OFF switch interface	1	pc	✓
33		6-pin JST wire, 2.54 mm for Driver control	3	pc	✓
34		6-core wire for Motor Driver Control, Verify 2m for each motor	6	m	✓
35		16 AWG red wire for 12 VDC supply	2	m	✓
36		16 AWG black wire for 12 VDC supply	2	m	✓
37		3-core mains cord for SMPS input	1	pc	✓
38		Male-to-male jumper wires	1	pack	✓
39		Male-to-female jumper wires	1	pack	✓
40		Female-to-female jumper wires	1	pack	✓
		41			
41	**Consumables**	Zip ties – 100 mm × 2.5 mm	1	pack	✓
42	**Tools**	Insulated screwdriver, combination (+) and (-)	1	pc	✓
43	**Consumables**	PTFE / Teflon tape	1	roll	✓
44	**Programming**	ESP32 programming cable	1	pc	✓
45	**Consumables**	Wire sleeves	1	pack	✓